

SOFTWARE DEVELOPMENT
CASE STUDY NO. 50

VoteCount

Election Result Dissemination System

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1. INTRODUCTION

VoteCount is a state election result dissemination system designed for a State Election Commission to handle live counting day operations. During a state election, vote counting takes place simultaneously across **4,100 counting tables** distributed over **70 constituencies**.

The main purpose of the VoteCount portal is to collect counting data entered at table terminals, verify the numbers against official records, allow digital sign-off by Returning Officers, and publish the verified results live to the general public.

Operational Scale & Key Problem

On counting day, the public web portal experiences massive user interest. The system is engineered to handle:

- **240,000 concurrent viewers** on the public portal.
- **18,000 web requests per second** during peak result updates.
- Synchronous result publication rounds **every 20 minutes**.

The primary technical and operational problem is maintaining **absolute accuracy and integrity** under heavy public traffic. In an election dissemination system, publishing an unverified, incorrect, or tampered result figure can lead to severe public panic and legal challenges.

Key Requirement: An unverified result must not be published.

The system must ensure that every published number is structurally verified, authenticated by a designated Returning Officer, and linked directly to an official physical Form 20 counting record before it can reach the public web portal.

2. PROJECT ESTIMATION AND STATUS

2.1 Project Data

Parameter	Value
Software Size	27 KLOC
Project Mode	Semi-detached
Available Window	7 months
BAC (Budget at Completion)	Rs. 1.05 crore
PV (Planned Value at Month 4)	Rs. 0.60 crore
EV (Earned Value at Month 4)	Rs. 0.51 crore
AC (Actual Cost at Month 4)	Rs. 0.66 crore
Concurrent Public Viewers	240,000
Web Request Rate	18,000 requests/sec
Publication Frequency	Every 20 minutes
Counting Tables	4,100 tables
Constituencies	70 constituencies

2.2 COCOMO Estimation

Software effort and duration are estimated using the **Basic COCOMO Model** for **Semi-detached** mode ($a = 3.0$, $b = 1.12$, $c = 2.5$, $d = 0.35$):

- **Effort (E):** $E = 3.0 \times (27)^{1.12} \approx 120.30$ person-months
- **Development Time (T):** $T = 2.5 \times (120.30)^{0.35} \approx 13.37$ months
- **Average Team Size:** $\text{Staff} = 120.30 / 13.37 \approx 9$ persons

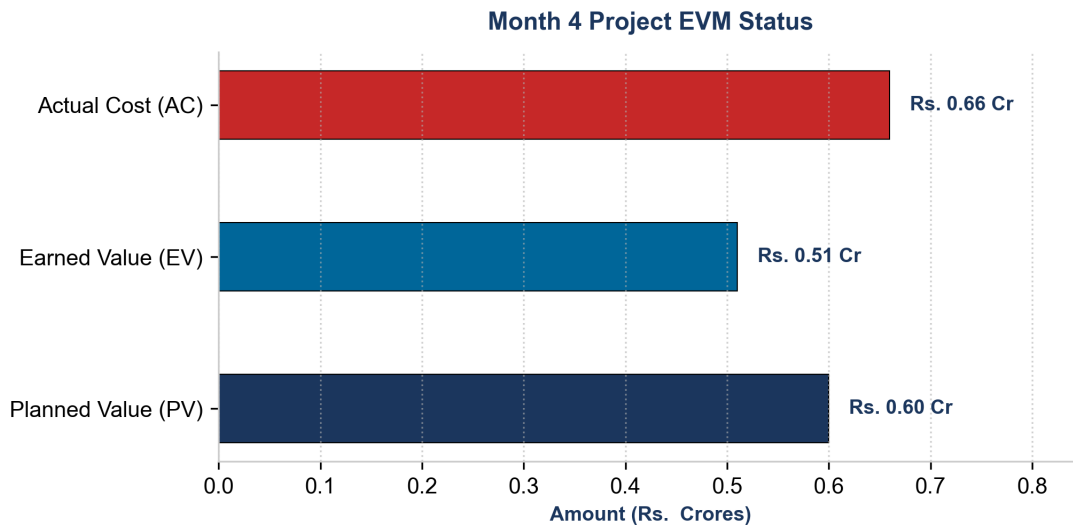
Parameter	Result
Project Size	27 KLOC
Mode	Semi-detached
Effort	120.30 PM
Development Time	13.37 months
Average Team	9 persons
Available Time	7 months
Difference	6.37 months

Conclusion: According to the COCOMO estimation, the nominal development time is higher than the available 7-month deadline. Therefore, parallel development, early testing and strict scope management are required.

2.3 Earned Value Management

At Month 4, EVM metrics are evaluated using: BAC = Rs. 1.05 Cr, PV = Rs. 0.60 Cr, EV = Rs. 0.51 Cr, AC = Rs. 0.66 Cr.

- **Schedule Variance (SV):** $EV - PV = 0.51 - 0.60 = -\text{Rs. } 0.09 \text{ crore}$ (Behind schedule)
- **Cost Variance (CV):** $EV - AC = 0.51 - 0.66 = -\text{Rs. } 0.15 \text{ crore}$ (Over budget)
- **Schedule Performance Index (SPI):** $EV / PV = 0.51 / 0.60 = 0.85$
- **Cost Performance Index (CPI):** $EV / AC = 0.51 / 0.66 \approx 0.773$
- **Estimate at Completion (EAC):** $BAC / CPI = 1.05 / 0.7727 \approx \text{Rs. } 1.359 \text{ crore}$
- **Variance at Completion (VAC):** $BAC - EAC \approx -\text{Rs. } 0.309 \text{ crore}$ ($-\text{Rs. } 30.88 \text{ lakh}$)
- **To-Complete Performance Index (TCPI):** $(BAC - EV) / (BAC - AC) = 0.54 / 0.39 \approx 1.385$



2.4 Project Status

- **Schedule:** Behind schedule — SPI = 0.85
- **Cost:** Over budget — CPI = 0.773
- **Forecast:** EAC = Rs. 1.359 Cr
- **Main Technical Concern:** Peak traffic management (240k concurrent viewers)
- **Main Quality Concern:** Unverified result publication
- **Action Plan:** Parallel development + early load testing + strict verification gate

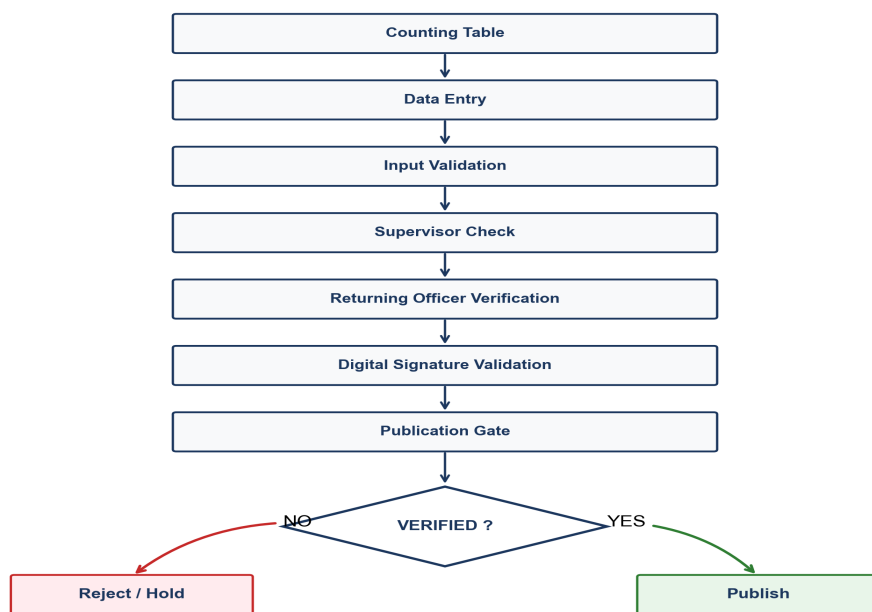
3. SOFTWARE REQUIREMENTS SPECIFICATION

3.1 Functional Requirements

Req ID	Title	Description	Priority
FR-01	Result Data Entry	Allow operators to enter table-wise vote counts for each 20-min round.	High
FR-02	Input Validation	Validate votes <= electors and check arithmetic sums.	High
FR-03	Supervisor Check	Supervisor reviews entered figures against physical Form 20.	High
FR-04	RO Verification	Returning Officer reviews aggregated constituency round totals.	High
FR-05	Digital Signature	Apply and validate RO's PKI digital signature before queuing.	High
FR-06	Publication Gate	Verify result status is VERIFIED before allowing publication.	High
FR-07	Result Release	Generate static JSON/HTML and push to CDN within 30s.	High
FR-08	Audit Logging	Log all state changes with timestamps and SHA-256 hashes.	High
FR-09	Correction Workflow	Post-publication corrections require formal RO re-verification.	High
FR-10	Round Mgmt	Track 20-minute publication rounds per constituency.	High
FR-11	MFA Auth	Enforce multi-factor authentication for official staff.	High
FR-12	Role Access	Enforce role-based access isolating operator rights to assigned tables.	High
FR-13	Result History	Maintain history of published rounds accessible to public users.	Med
FR-14	Offline Buffering	Buffer data locally during network outage; auto-sync on reconnect.	High

3.2 Result Verification Flow

Figure 1: Result Verification Flow

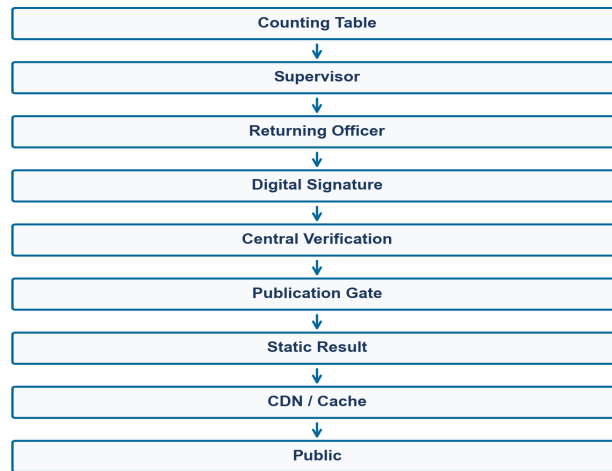


3.3 Non-Functional Requirements

- **Performance & Load:** Expected peak load of **240,000 concurrent viewers** and **18,000 requests/sec**. Stress target is **360,000 concurrent viewers** and **27,000 requests/sec**.
- **Security:** TLS 1.3 encryption, AES-256 storage, PKI digital signatures for Returning Officer sign-off.
- **Availability:** 99.99% system uptime during the counting day period.
- **Integrity:** Zero modification of published data without explicit RO re-verification and audit logging.

3.4 Verification Chain

Verification Chain Flowchart



Counting data is entered at table terminals, reviewed by supervisors against physical Form 20 forms, approved by Returning Officers with digital signatures, validated by central verification rules, passed through the publication gate, converted into static JSON/HTML, cached on CDN edge nodes, and displayed to public users.

3.5 Publication Rule

Key Requirement: An unverified result must not be published.

```
IF (Result.Status == "VERIFIED")  
AND (Digital_Signature == "VALID")  
AND (Validation_Rules == "PASSED")  
THEN Publish Result to CDN  
ELSE Reject / Hold Record
```

3.6 Audit and Traceability

Every figure published on the web portal must map directly back to its origin:

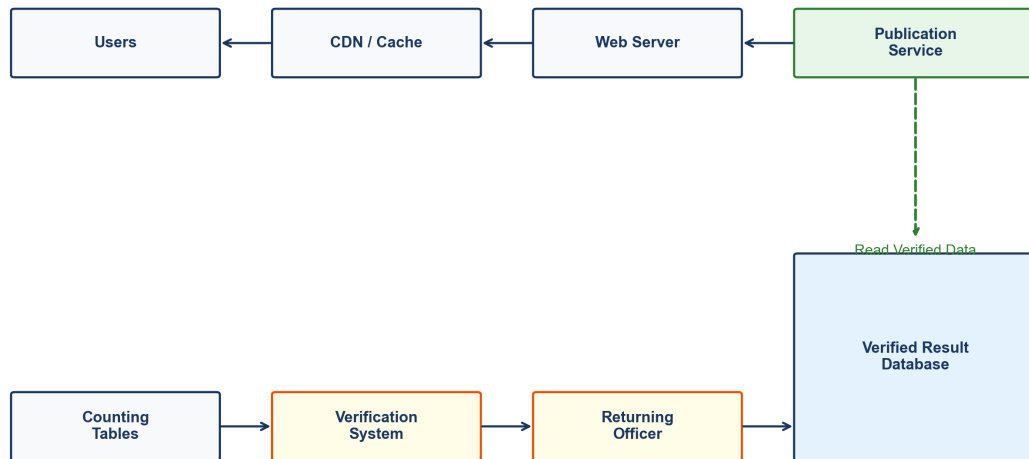
Public Figure → Publication ID → Round Number → Counting Table → Signed Form 20 → RO Signature

This unbroken chain guarantees full accountability and auditability for election audit teams.

4. SYSTEM DESIGN

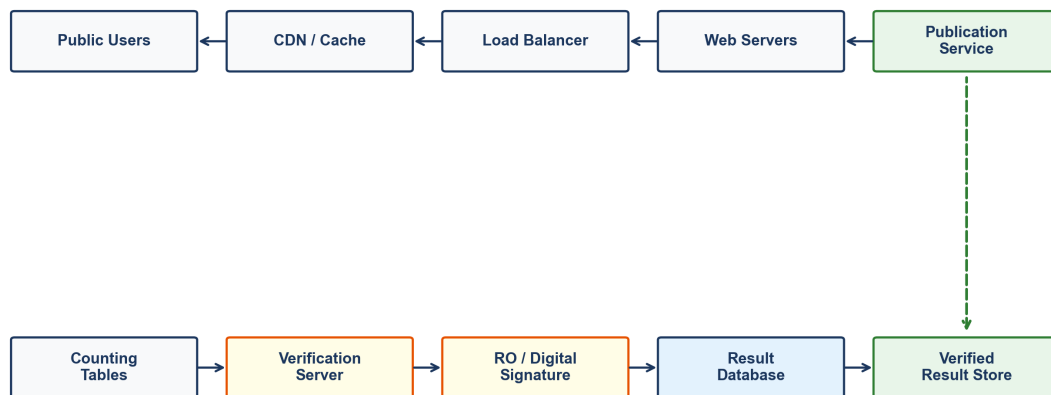
4.1 System Architecture

System Architecture Diagram



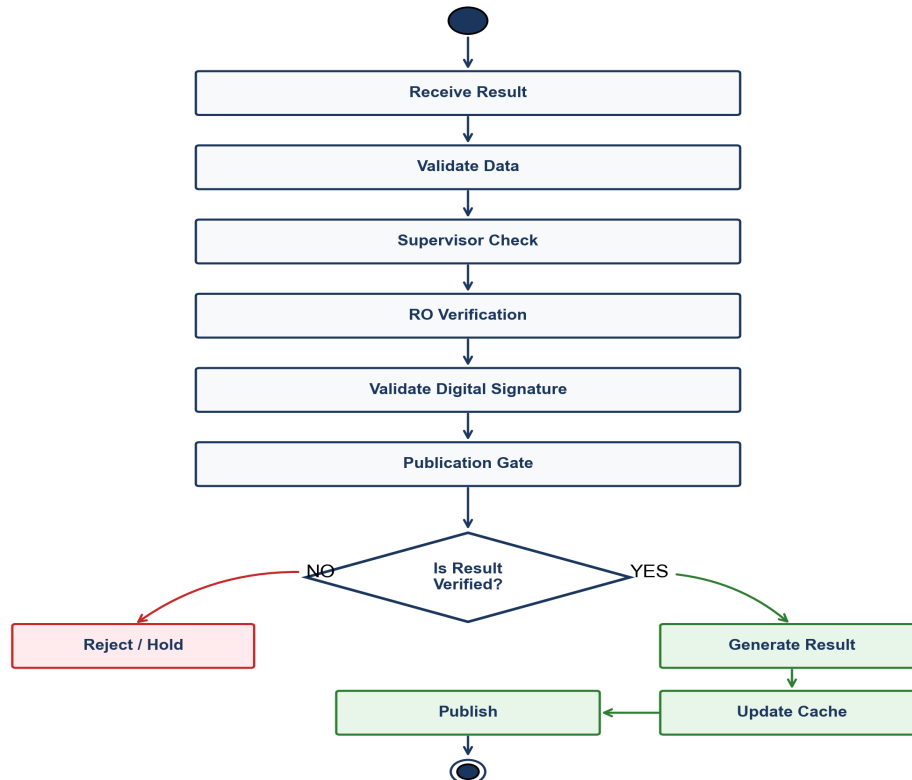
4.2 Deployment Diagram

Deployment Diagram



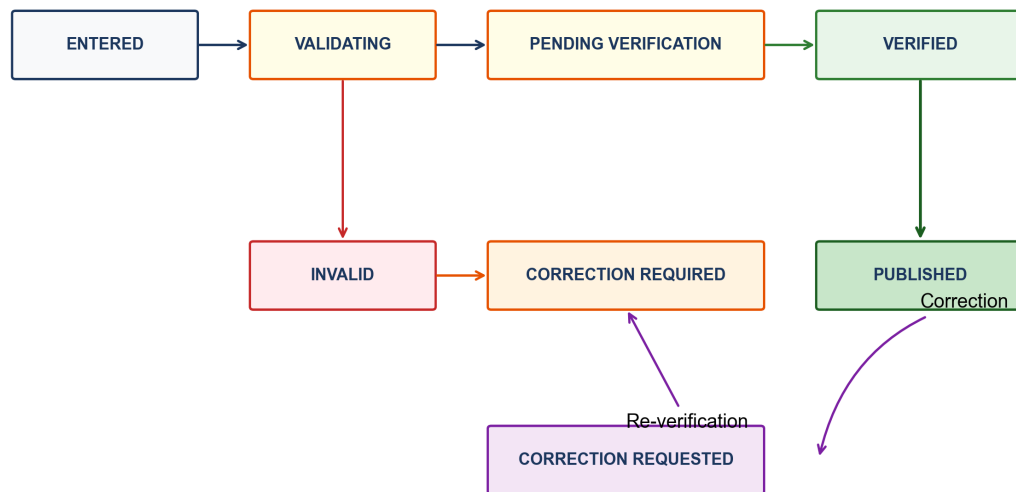
4.3 Activity Diagram

Publication Round Activity Diagram



4.4 Result State Diagram

Result State Diagram



State Transitions Summary

- **Normal Flow:** ENTERED → VALIDATING → PENDING VERIFICATION → VERIFIED → PUBLISHED
- **Error Path:** VALIDATING → INVALID → CORRECTION REQUIRED → ENTERED
- **Post-Release Correction:** PUBLISHED → CORRECTION REQUESTED → RE-VERIFICATION → VERIFIED → PUBLISHED

5. TESTING AND REHEARSAL PLAN

5.1 Testing Strategy

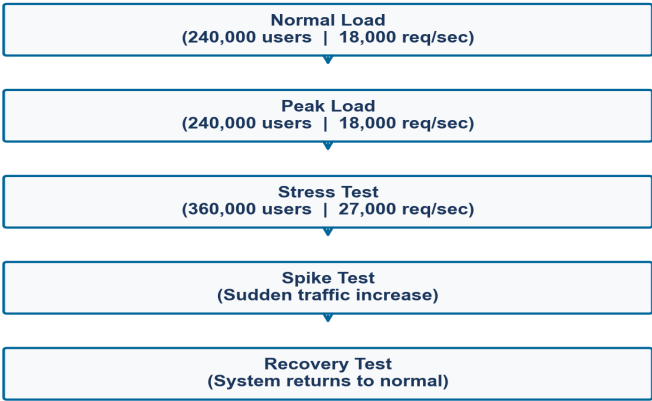
Testing focuses on functional accuracy across all 4,100 counting tables, cryptographic validation of digital signatures, load capacity under peak public traffic, and failure recovery.

5.2 Load Testing

Load testing is conducted at **1.5 times expected peak volume**:

- Expected Viewers: 240,000 → **360,000 Concurrent Viewers (1.5x Target)**
- Expected Throughput: 18,000 req/sec → **27,000 Requests/Second (1.5x Target)**

Load Testing Levels



Metric	Acceptance Target	Failure Threshold
Static Page Response Latency	< 800 ms	> 1,500 ms
Error Rate (HTTP 5xx)	< 0.01%	> 0.10%
Throughput Capacity	>= 27,000 req/sec	< 25,000 req/sec
CDN Cache Hit Ratio	> 99.5%	< 98.0%

5.3 Verification Testing

Test ID	Scenario	Expected Outcome
VT-01	Valid signed result	Status set to PUBLISHED; CDN cache updated.
VT-02	Missing signature	Blocked at Publication Gate; status remains PENDING.
VT-03	Invalid / tampered signature	Hard rejection at Publication Gate; security alert raised.
VT-04	Unverified result submission	Rejected by Publication Gate; zero DB read by public API.
VT-05	Arithmetic sum mismatch	Form validation error thrown; entry blocked.
VT-06	Corrected result re-submission	Old record marked SUPERSEDED; requires RO re-signing.

5.4 Full Dress Rehearsal

A full dress rehearsal is conducted at **Month 6.8** involving all 70 constituency centers, table operators, supervisors, and Returning Officers. The rehearsal simulates full counting operations, network outages, and 360,000 user load bursts to confirm go-live readiness.

6. RISK MANAGEMENT

6.1 Risk Exposure

Risk exposure is calculated as: **Exposure = Probability (P) × Impact (I)**

Risk	P	I	Exposure (E)
Portal Collapse under Load	0.5	9	4.5
Counting Table Data Entry Error	0.6	7	4.2
Network Outage at Constituency	0.4	6	2.4
Unverified Figure Publication	0.2	10	2.0

Note: Although Unverified Figure Publication has an exposure score of 2.0 due to lower probability, its **Impact score is 10 (Maximum Severity)**, making it a top priority for technical controls.

6.2 RMMM

Risk	Mitigation	Monitoring	Contingency
Portal Collapse	CDN caching + load testing	Server metrics & latency	Static backup landing page
Data Error	Validation + supervisor check	Input error logs	Manual Form 20 check
Network Outage	Offline buffering	Network status pings	Manual phone/fax transmission
Unverified Result	Publication Gate	Audit log monitor	Freeze publication portal

6.3 Manual Fallback

- **Portal Failure:** Direct public traffic to static mirror landing pages and broadcast official updates via media feeds.
- **Data Error:** Compare digital entry against physical signed Form 20 hardcopy.
- **Network Link Down:** Use secure voice/fax communication to relay verified totals to central HQ.
- **Unverified Release Alert:** Immediately freeze public updates and issue official State Election Commission press clarification.

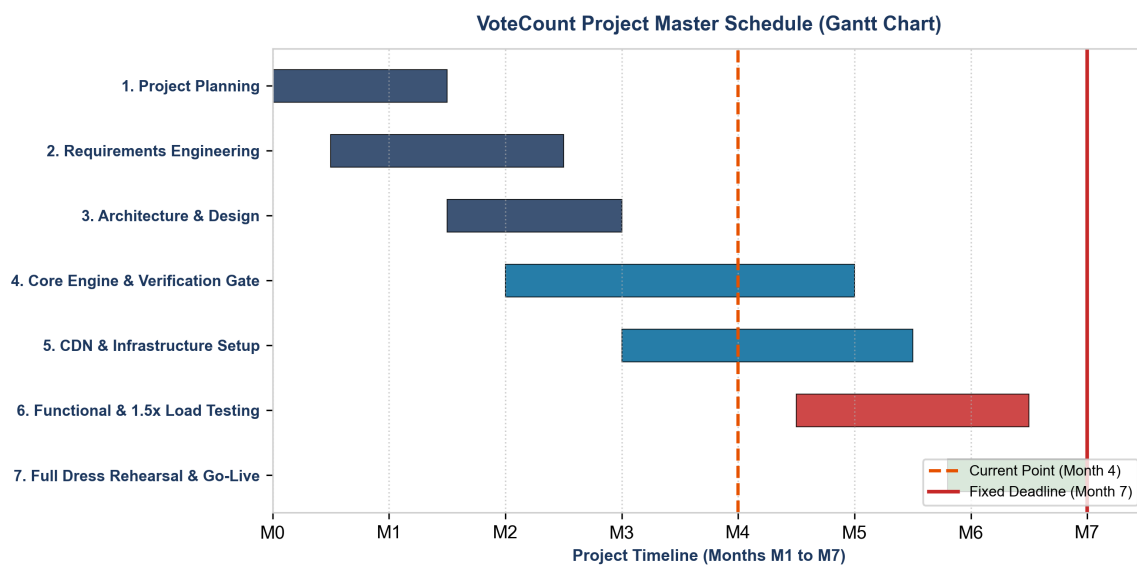
7. PROJECT MANAGEMENT PLAN

7.1 WBS

1.0 VoteCount System Project

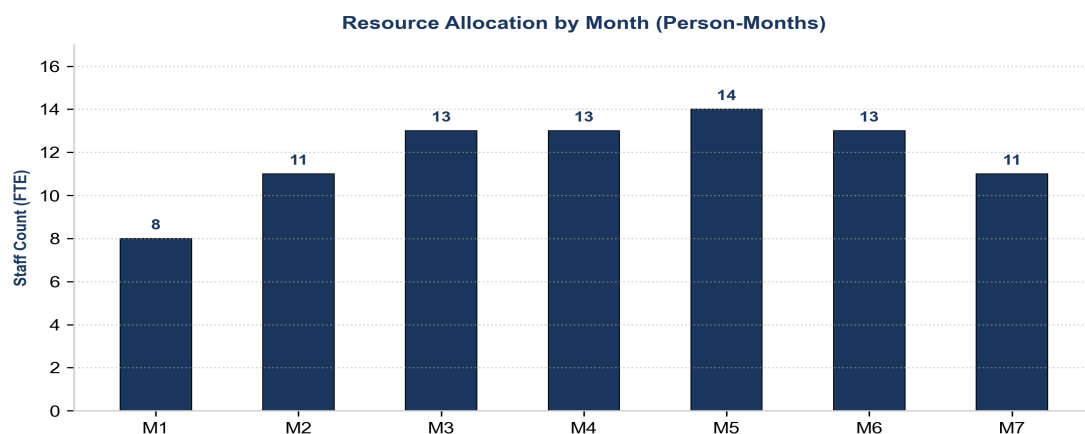
- 1.1 Project Management & EVM Control
- 1.2 Software Requirements Specification (SRS)
- 1.3 System Architecture & Gate Design
- 1.4 Core Development (Data Entry, Verification, Static Engine)
- 1.5 CDN Infrastructure & Database Hardening
- 1.6 Functional, Security & 1.5x Load Testing
- 1.7 Full 70-Constituency Dress Rehearsal
- 1.8 Production Deployment & Support

7.2 Gantt Chart



7.3 Resource Allocation by Month

Role	M1	M2	M3	M4	M5	M6	M7
Project Manager	1	1	1	1	1	1	1
Requirements Analyst	2	1	1	0	0	0	0
Software Developers	3	4	5	5	5	3	2
QA / Test Engineers	1	1	2	3	4	4	3
DevOps / Infrastructure	1	2	2	2	2	3	3
Security Engineer	0	1	1	1	1	1	1
DBA / Data Engineer	0	1	1	1	1	1	1
Total Staff (FTE)	8	11	13	13	14	13	11



7.4 Scope Reduction

Priority	Features
Must Have	Verification, security, audit, result publication, data validation.
Can Simplify	UI design, dashboards, visualisations.
Can Defer	Advanced analytics, optional reports, extra notifications.

The project deadline cannot move, so optional features will be reduced before any core verification or security feature.

8. CONCLUSION

1. **COCOMO Estimation:** Nominal effort is 120.30 person-months, requiring 13.37 months nominal duration.
2. **Schedule Constraint:** The fixed 7-month deadline requires fast-tracking and parallel execution.
3. **EVM Status:** Month 4 indicators (SPI = 0.85, CPI = 0.773) show schedule slippage and cost overrun, forecasting EAC of Rs. 1.359 crore.
4. **Architecture Solution:** Static page generation and CDN edge caching handle 240,000 concurrent viewers (and 360,000 stress viewers) while insulating backend databases.
5. **Quality Gate:** Automated Publication Gate enforces the requirement that **an unverified result must not be published**.
6. **Testing & Rehearsal:** 1.5x stress testing (27,000 req/sec) and a 70-constituency dress rehearsal at Month 6.8 ensure operational readiness.
7. **Risk Strategy:** Risk exposure prioritization ensures high-impact issues have tested manual fallbacks.
8. **Scope Discipline:** Core verification features are protected above all secondary visual features.

REFERENCES / APPENDIX

- **COCOMO Baseline:** $a = 3.0$, $b = 1.12$, $c = 2.5$, $d = 0.35$ (Semi-detached mode).
- **EVM Baseline:** BAC = Rs. 1.05 Cr, PV = Rs. 0.60 Cr, EV = Rs. 0.51 Cr, AC = Rs. 0.66 Cr.
- **Test Case Summary:** Key functional, verification and load test cases are listed in the testing section.
- **Traceability Mapping:** All public figures map directly to signed Form 20 physical forms.